

4-12 Find R_{IN} in Figure P4-12.

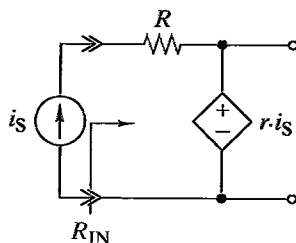


FIGURE P4-12

4-13 Find R_{IN} in Figure P4-13.

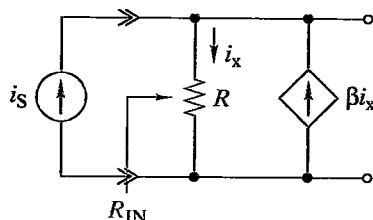


FIGURE P4-13

4-14 Find the Thévenin equivalent circuit seen by the load in Figure P4-14.

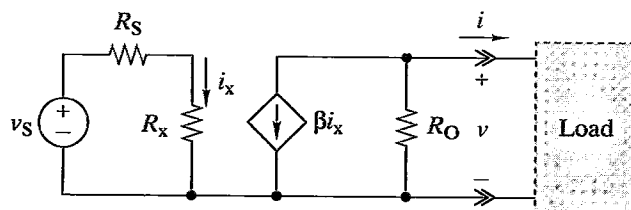


FIGURE P4-14

4-15 Find the Norton equivalent circuit seen by the load in Figure P4-15.

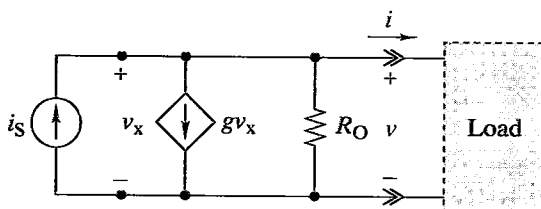


FIGURE P4-15

4-16 The circuit parameters in Figure P4-16 are $R_B = 3 \text{ k}\Omega$, $R_C = 4 \text{ k}\Omega$, $\beta = 120$, $V_\gamma = 0.7 \text{ V}$, and $V_{CC} = 15 \text{ V}$. Find v_{CE} for $v_S = 2 \text{ V}$. Repeat for $v_S = 5 \text{ V}$.

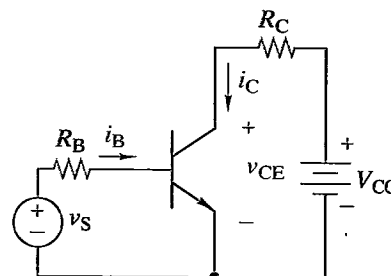


FIGURE P4-16

4-17 The circuit parameters in Figure P4-16 are $R_C = 3 \text{ k}\Omega$, $\beta = 100$, $V_\gamma = 0.7 \text{ V}$, and $V_{CC} = 5 \text{ V}$. Select a value of R_B such that the transistor is in saturation mode and $v_S \geq 2 \text{ V}$.

4-18 The parameters of the transistor in Figure P4-18 are $\beta = 50$ and $V_\gamma = 0.7 \text{ V}$. Find i_C and v_{CE} for $v_S = 0.8 \text{ V}$. Repeat for $v_S = 2 \text{ V}$.

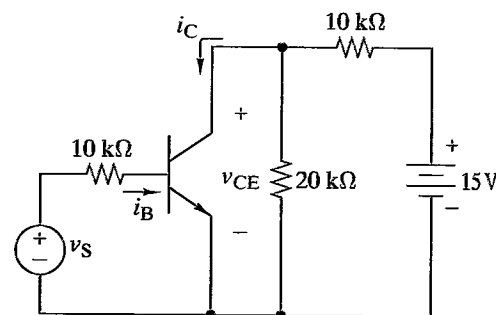


FIGURE P4-18

OBJECTIVE 4-2 OP AMP CIRCUIT ANALYSIS (SECTS. 4-3, 4-4)

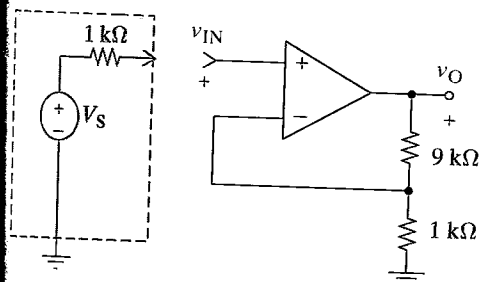
Given a circuit consisting of linear resistors, OP AMPs, and independent sources, find selected output signals or input-output relationships.

See Examples 4-13, 4-14, 4-16, 4-17, 4-18 and Exercises 4-14, 4-15, 4-17, 4-18, 4-19, 4-20, 4-21

4-19 Two OP AMP circuits are shown in Figure P4-19. Both claim to produce a gain of either $+10$ or -10 .

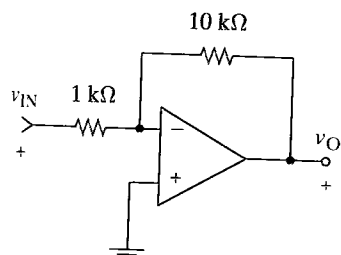
(a) Show that the claim is true.

(b) A practical source with a series resistor of $1 \text{ k}\Omega$ is connected to the input of each circuit. Does the original claim still hold? If it does not, explain why.



Source

Circuit 1



Circuit 2

FIGURE P4-19

Suppose the output of the practical source shown in Figure P4-19 needs to be amplified by -100 and you can use only the two circuits shown. How would you connect the circuits to achieve this? Explain.

(a) Find v_O in terms of v_S in Figure P4-21.

(b) Validate your answer by simulating the circuit in OrCAD.

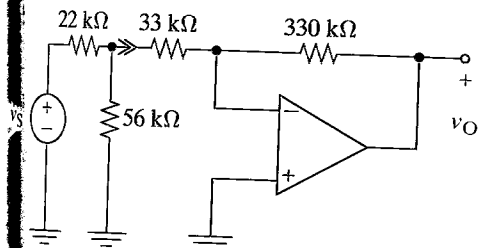


FIGURE P4-21

What is the range of the gain v_O/v_S in Figure P4-22?

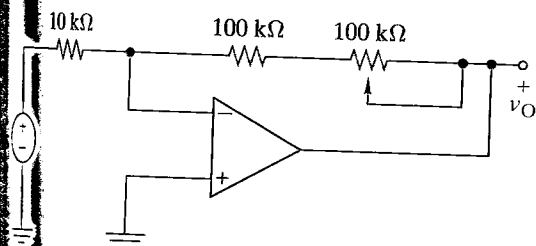


FIGURE P4-22

4-23 (a) Find v_O in terms of v_S in Figure P4-23.

(b) Find i_O for $v_S = 1$ V. Repeat for $v_S = 3$ V.

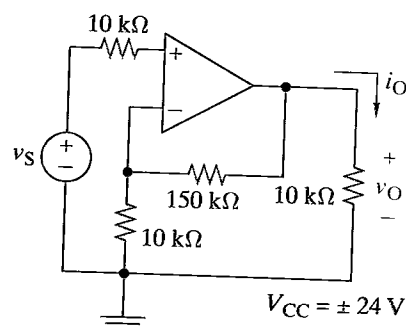


FIGURE P4-23

4-24 (a) Find v_O in terms of v_S in Figure P4-24.

(b) Find i_O for $v_S = 2$ V.

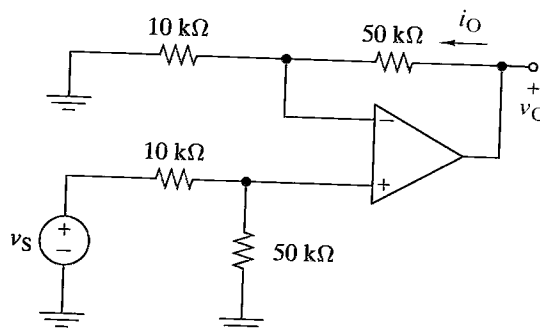


FIGURE P4-24

4-25 What is the gain v_O/v_S in Figure P4-25 when the switch is in position 1? Repeat for positions 2 and 3.

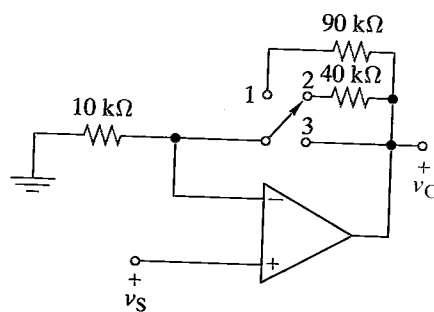


FIGURE P4-25

4-26 (a) Find v_O in terms of the inputs v_1 and v_2 in Figure P4-26.

- (b) If $v_1 = 1$ V, what is the range of values v_2 can have without saturating the OP AMP?

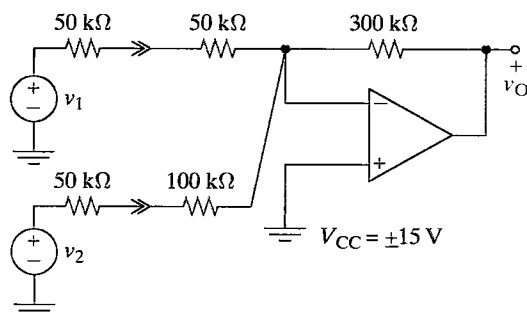


FIGURE P4-26

- 4-27 The input-output relationship for a three-input inverting summer is

$$v_O = -[v_1 + 10v_2 + 100v_3]$$

The resistance of the feedback resistor is $100\text{ k}\Omega$. Find the values of the input resistors R_1 , R_2 , and R_3 .

- 4-28 Find v_O in terms of the inputs v_1 and v_2 in Figure P4-28.

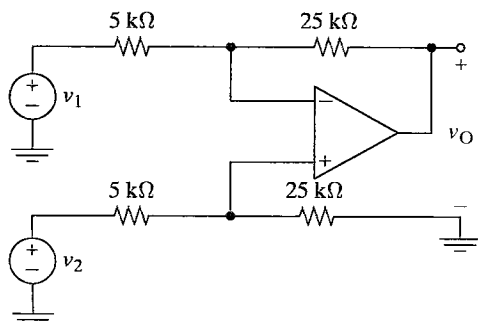


FIGURE P4-28

- 4-29 The switch in Figure P4-29 is open. Find v_O in terms of the inputs v_{S1} and v_{S2} . Repeat with the switch closed.

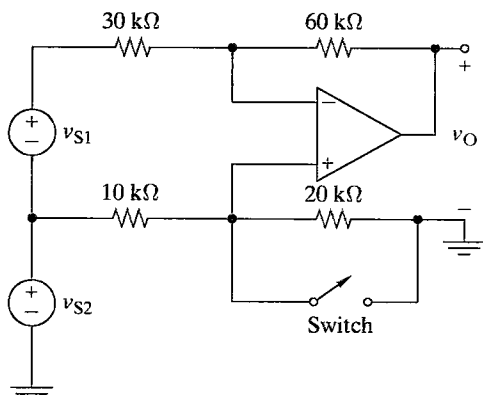


FIGURE P4-29

- 4-30 Find v_O in terms of v_{S1} and v_{S2} in Figure P4-30.

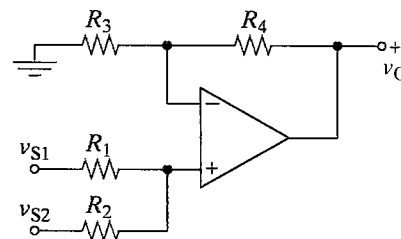


FIGURE P4-30

- 4-31 It is claimed that $v_O = v_S$ when the switch in Figure P4-31 is closed and that $v_O = -v_S$ when the switch is open. Prove or disprove this claim.

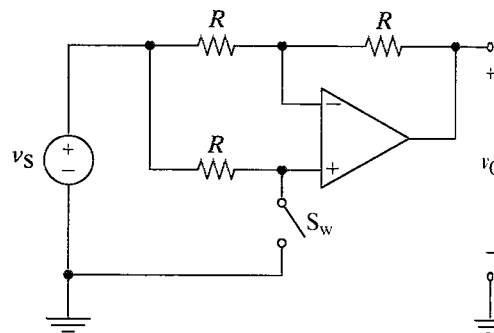


FIGURE P4-31

- 4-32 The circuit in Figure P4-32 has a diode in its feedback path and is called a "log amp" because its output is proportional to the natural log of the input.

(a) Show that $v_O = -V_T \ln(1 + v_S/(R_S I_O))$ if the i - v characteristics of the diode is $i_D = I_O(e^{v_D/V_T} - 1)$.

(b) Using MATLAB, plot v_O versus v_S for $R_S = 10\text{ k}\Omega$, $I_O = 10^{-14}$ A, and $V_T = 0.025$ V. Plot your results on a semilog plot for $10^{-6} \text{ V} \leq v_S \leq 100 \text{ V}$.

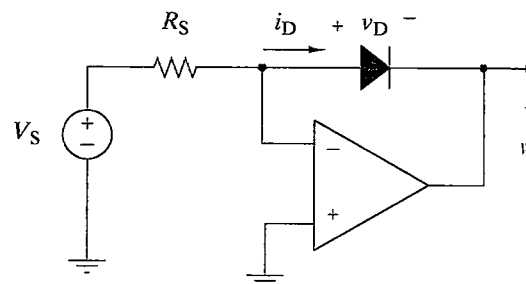


FIGURE P4-32

- 4-33 (a) Use node voltage analysis to find the input-output relationship or K of the circuit in Figure P4-33.

- (b) Select values for the resistors so that $K = -3$.

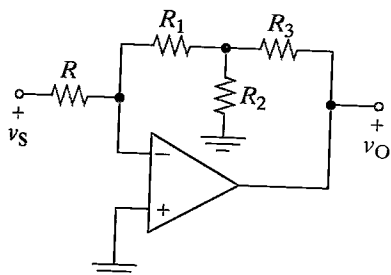


FIGURE P4-33

- 4-34 Use node voltage analysis in Figure P4-34 to show that $i_o = -v_s/2R$ regardless of the load. That is, show that the circuit is a voltage-controlled current source.

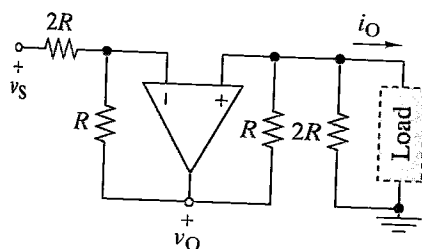


FIGURE P4-34

- 4-35 Find v_o in terms of the inputs v_{s1} and v_{s2} in Figure P4-35.

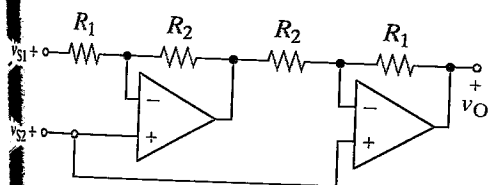


FIGURE P4-35

- 4-36(a) Find v_o in terms of the inputs v_{s1} and v_{s2} in Figure P4-36.

- (b) Redesign the circuit using only one OP AMP.

- (c) Validate your design using OrCAD.

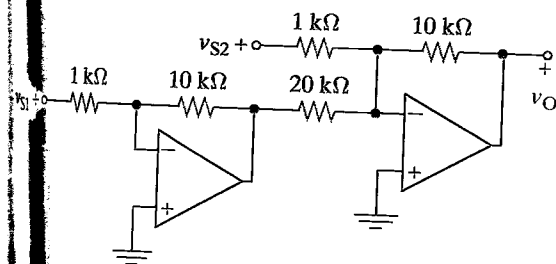
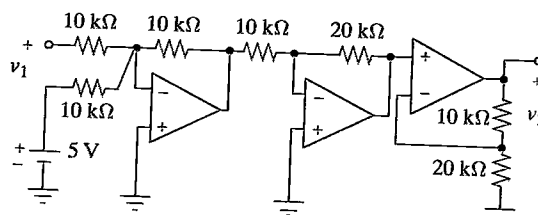
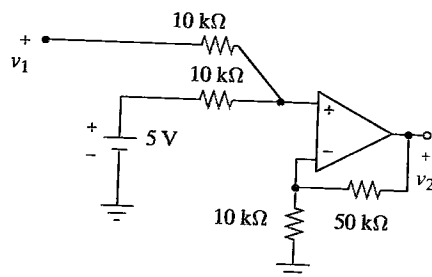


FIGURE P4-36

- 4-37 On an exam, students were asked to design an efficient solution for the following relationship: $v_2 = 3v_1 + 15$. Two of the designs are shown in Figure P4-37. Which, if either, of the designs are correct and what grade would you award each student?



(a)



(b)

FIGURE P4-37

- 4-38 Find the output v_2 in terms of the input v_1 in Figure P4-38.

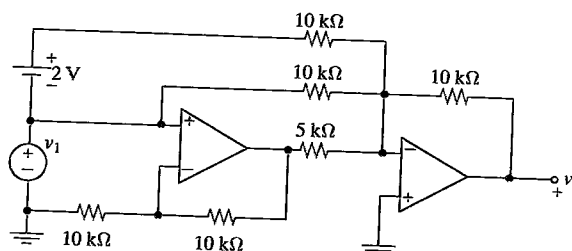


FIGURE P4-38

- 4-39 Find the output v_o in terms of the input v_s in Figure P4-39.

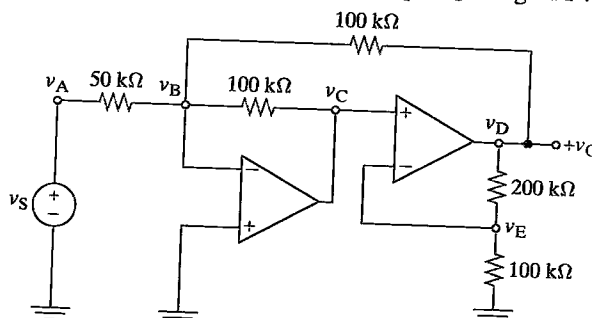


FIGURE P4-39

- 4-40 Find the output v_O in terms of the input v_1 in Figure P4-40. Verify your answer using OrCAD.

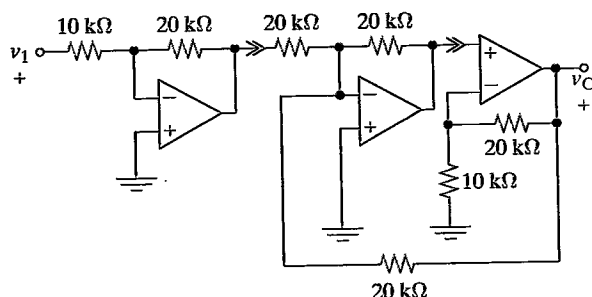


FIGURE P4-40

OBJECTIVE 4-3 OP AMP CIRCUIT DESIGN AND EVALUATION (SECT. 4-5)

Given an input-output relationship, use resistors and OP AMPs to design one or more circuits that implement the relationship within the stated constraints. Choose the better of two working circuits to meet certain constraints.

See Design Examples 4-12, 4-15, 4-20, 4-21, 4-22, 4-23, 4-24, Design Exercises 4-12, 4-16, 4-22, 4-28, 4-29, 4-30, Evaluation Examples 4-19, 4-20, 4-21, 4-22

- 4-41 Design an OP AMP amplifier with a voltage gain of -10 and an input resistance greater than $10\text{ k}\Omega$ using standard 5% resistance values less than $300\text{ k}\Omega$.
- 4-42 Design an OP AMP amplifier with a voltage gain of 4 using only $10\text{-k}\Omega$ resistors and one OP AMP.
- 4-43 Using a single OP AMP, design a circuit with inputs v_1 and v_2 and an output $v_O = 2v_2 - 3v_1$. The input resistance seen by each input should be greater than $5\text{ k}\Omega$.
- 4-44 Design a differential amplifier with inputs v_1 and v_2 and an output $v_O = 10(v_2 - v_1)$ using only one OP AMP. All resistances must be between $5\text{ k}\Omega$ and $200\text{ k}\Omega$.
- 4-45 Using no more than two OP AMPs, design an OP AMP circuit with inputs v_1 , v_2 , and v_3 and an output $v_O = -3v_1 + 2v_2 - 5v_3$.
- 4-46 Design the interface circuit in Figure P4-46 so that 10 mW is delivered to the $100\text{-}\Omega$ load. Repeat for a $100\text{-k}\Omega$ load. Verify your designs using OrCAD.

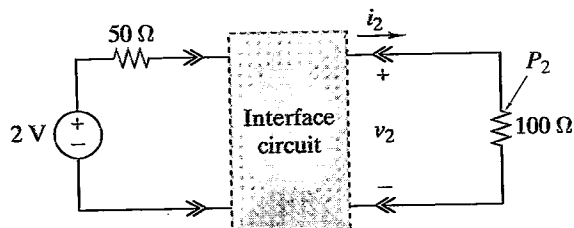


FIGURE P4-46

- 4-47 Design the interface circuit in Figure P4-47 so that the output is $v_2 = 150v_1 + 1.5\text{ V}$.

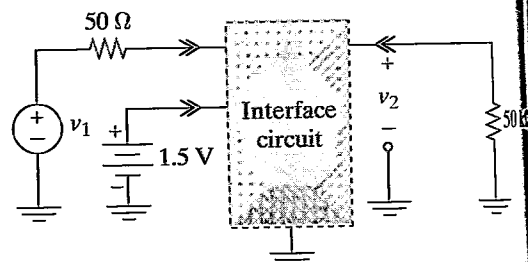
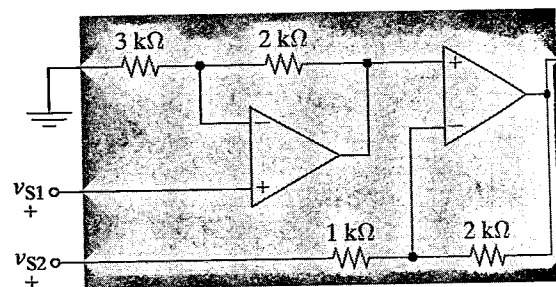


FIGURE P4-47

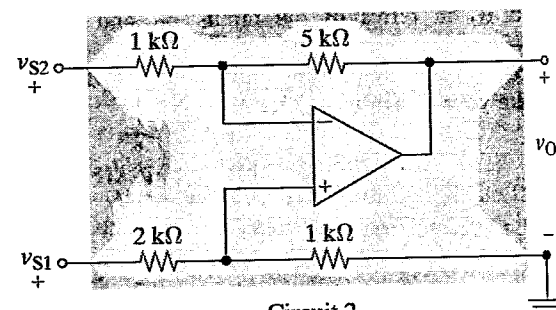
- 4-48 (a) Design a circuit that can produce $v_O = 100v_1 + 5\text{ V}$ using two OP AMPs.
(b) Repeat using only one OP AMP.
- 4-49 A requirement exists for an OP AMP circuit with the input-output relationship

$$v_O = 5v_{S1} - 2v_{S2}$$

Two proposed designs are shown in Figure P4-49. As a project engineer, you must recommend one of these circuits for production. Which of these circuits would you recommend for production and why? (Hint: First verify that the circuit performs the required function.)



Circuit 1



Circuit 2

FIGURE P4-49

- 4-50 A requirement exists for an OP AMP circuit that deliver 12 V to a $1\text{-k}\Omega$ load using a 4-V source as an input.

- 4-56 **D** A small pressure transducer has the characteristics shown in Figure P4-56. Design an interface that will operate between 7 and 32 psi. An input of 7 psi should produce -5 V and 32 psi should produce 5 V. The transducer is modeled as a voltage source in series with a $500\ \Omega$ resistor that can vary by $\pm 75\ \Omega$ depending on the pressure. The OP AMPs you must use have a maximum closed-loop gain of 2000. Your available bias source is 5 V.

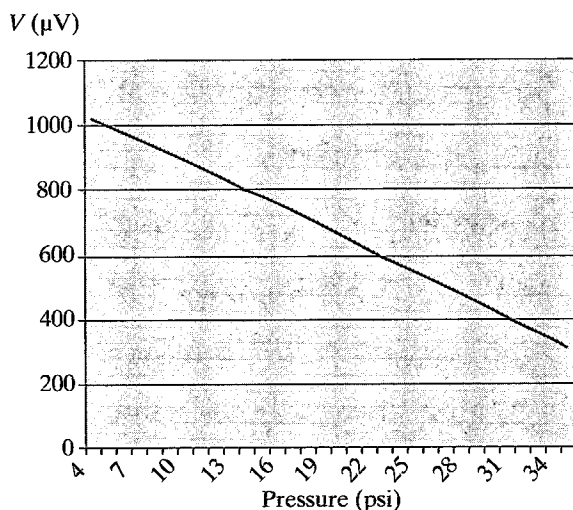
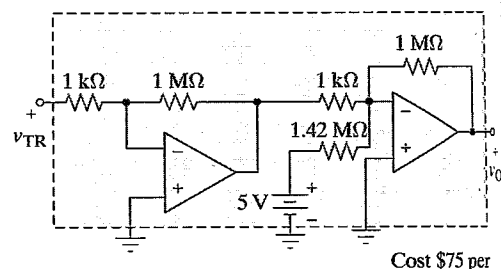


FIGURE P4-56

- 4-57 **D** A medical-grade pressure transducer has been developed for use in invasive blood pressure monitoring. The output voltage of the transducer is $v_{TR} = (0.05P - 0.75)$ mV, where P is pressure in mm Hg. The output resistance of the transducer is $300\ \Omega$. The blood pressure measurement is to be an input to an existing multisensor monitoring system. This system treats a 1 V input as a blood pressure of 20 mm Hg and a 10 V input as 200 mm Hg. Design an OP AMP circuit to interface the new pressure transducer with the existing monitoring system.
- 4-58 **D** The acid/alkaline balance of a fluid is measured by the pH scale. The scale runs from 0 (extremely acid) to 14 (extremely alkaline), with pH 7 being neutral. A pH electrode is a sensor that produces a small voltage that is directly proportional to the pH of the fluid in a test chamber. For a certain pH electrode, the proportionality factor is 60 mV/pH. A preamplifier is needed to interface this sensor with a variety of laboratory instruments. The output of the preamp must be 1 V when the sensor is immersed in a test solution with pH = 4 and 1.75 V when it is immersed in a solution with pH = 7. Design an amplifier to meet these requirements.
- 4-59 **D** A photoresistor varies from $100\ \Omega$ in bright sunlight to $200\ \text{k}\Omega$ in total darkness. Design a suitable circuit using the photoresistor so that total darkness produces 5 V while bright sunlight produces -5 V, regardless of the load. You have a 5 -V source and a ± 15 -V source to power any OP AMP you may need.

- 4-60 **E** Your engineering firm needs an instrumentation amplifier that provides the following input-output relation: $v_O = 10^6 v_{TR} - 3.5$ V. The transducer is modeled as a voltage source in series with a resistor that varies with transducer voltage from $40\ \Omega$ to $750\ \Omega$. A vendor is offering the amplifier shown in Figure P4-60 and agrees to make a single change to the amplifier, if needed, for no cost. Would you recommend buying it? Explain your rationale.



Instrumentation Amplifier Model 54

FIGURE P4-60

- 4-61 The OP AMP in Figure P4-61 operates as a comparator. Find the output voltage when $v_S = 2$ V. Repeat for -2 V and $v_S = 6$ V.

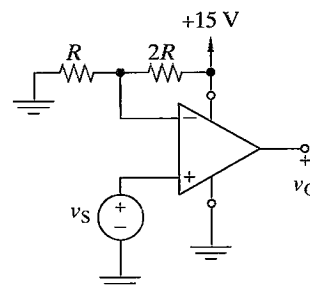


FIGURE P4-61

- 4-62 The circuit in Figure P4-62 has $V_{CC} = 15$ V and $v_N = -2$ V. Sketch the output voltage v_O on the range $0 \leq t \leq 2$ s for $v_S(t) = 10 \sin(2\pi t)$ V.

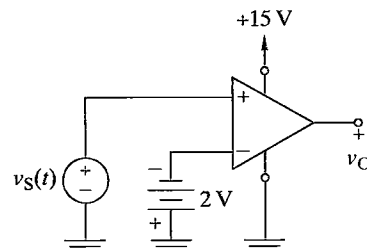


FIGURE P4-62

- 4-63 **D** A signal varies as $v_S(t) = 10(1 - e^{-100t})$ V. Design a comparator circuit that will switch from 0 to 5 V in 2 ms.